

#06 AWS S3 to Google Drive Intelligent Sync

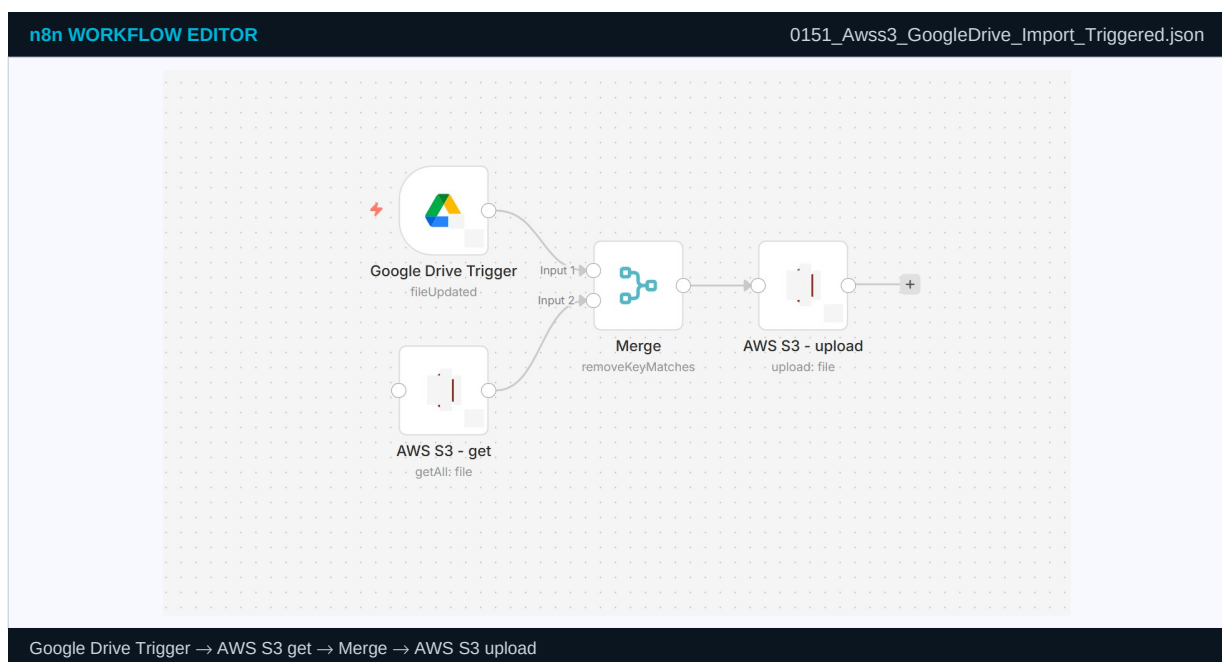
AWS / Cloud
Storage
Automation

Bidirectional AWS S3 and Google Drive sync with automatic file routing

BUSINESS PROBLEM

Engineering teams use S3 while business teams use Google Drive. Files constantly get lost between systems. This workflow watches Drive for new files, syncs to the correct S3 bucket with intelligent routing by folder and file type, keeping both platforms in sync automatically.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Google Drive Trigger — file created or updated event

```
{
  "fileId":
  "1BxiMVs0XRA5nFMdKvBdBZjgmUUqptlbs74OgVE2upms",
  "fileName": "Q2-2024-Operations-Report.pdf",
  "mimeType": "application/pdf",
  "size": 862144,
  "parentFolder": "Finance/Reports/2024/Q2",
  "modifiedTime": "2024-06-15T08:30:00Z",
  "modifiedBy": "sarah@company.com"
}
```

OUTPUT — AWS S3 upload result — file stored in correct bucket path

```
{
  "json": {
    "synced_file": "Q2-2024-Operations-Report.pdf",
    "source": "Google Drive /Finance/Reports/2024/Q2",
    "s3_key":
    "finance/reports/2024/06/Q2-2024-Operations-Report.pdf",
    "bucket": "company-docs-prod",
    "etag": "d41d8cd98f00b204e9800998ecf8427e",
    "size_bytes": 862144,
    "content_type": "application/pdf",
    "synced_at": "2024-06-15T08:30:44Z",
    "duration_ms": 1840
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Google Drive Trigger — file created	Watches target Drive folder. Fires on new or modified file: {fileId, fileName, mimeType, size, parentFolder, modifiedTime}.
2	■ AWS S3 get — fetch existing file list	Lists current S3 bucket contents for duplicate detection before upload. Prevents re-uploading unchanged files.
3	■ Merge — join metadata and file binary	Merge node combines Drive trigger metadata with downloaded binary content into single workflow item for S3 upload.
4	■ AWS S3 upload — intelligent routing	Key computed from folder path: {department}/{year}/{month}/{filename}. ContentType set from mimeType. Returns ETag.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
Google Drive OAuth2	console.cloud.google.com → Drive API v3 → Create OAuth2 credentials	Google Drive OAuth2
AWS Access Key + Secret	AWS Console → IAM → Policy: s3:PutObject s3:GetObject on target bucket	AWS Credentials

SAMPLE RUN — what this workflow processed and produced

Scenario	An engineering consultancy stores project documents in Google Drive for collaboration but needs all files mirrored to AWS S3 for backup and integration with their analytics pipeline.	
Input received	Sarah from finance saved the Q2 Operations Report PDF to the Drive Finance/Reports/2024/Q2 folder at 08:30. The Drive trigger fired within 4 seconds of the save event.	
What the workflow found	S3 bucket check showed no existing file with that name in the finance/reports/2024/06/ path. File was a new document, not an update. Routing logic mapped /Finance/ parent folder to the finance/ S3 prefix.	
Output produced	PDF uploaded to s3://company-docs-prod/finance/reports/2024/06/ in 1.84 seconds. ETag confirmed successful upload. Analytics pipeline automatically detected new file in S3 and included it in the next nightly report run.	
JSON: 0151_Awss3_GoogleDrive_Import_Triggered.json	GitHub: trm-s3-drive-intelligent-sync	aws.amazon.com/s3/